SIMATS ENGINEERING

SIMATS Online Programming Lab

C++ PROGRAMMING

MANNE TEJASWINI
192372285RA

CO4_AT1_SIMATS_Compiler

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QUESTIONS

Q1 ✓
Multiple Inheritance in Smart Buildings
10 marks

Q2 ✓
Hierarchical Inheritance for EVs
10 marks

Q3 ✓
Function Overriding in Green Transport
10 marks

Q4 ✓
Multilevel Inheritance for Circular Economy
10 marks

Q5 ✓
Abstract Class for Carbon Footprint
10 marks

5/5 answered

Q5 Abstract Class for Carbon Footprint

10 marks

A carbon-footprint calculator calculates emissions for different activities such as Transport and Electricity. Analyze how abstract classes and virtual functions ensure flexibility.

Your Code

```cpp
#include <iostream>
using namespace std;

class CarbonFootprint {
public:
    virtual void calculateEmission() = 0;
};

class Transport : public CarbonFootprint {
public:
    void calculateEmission() override {
        cout << "Transport Emission: 120 kg CO2" << endl;
    }
};

class Electricity : public CarbonFootprint {
public:
```

Input

stdin input

Output

Visible Test Cases

Test Case 1

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QUESTIONS

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- **Q2** Hierarchical Inheritance for EVs — 10 marks
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- **Q4** Multilevel Inheritance for Circular Economy — 10 marks
- **Q5** Abstract Class for Carbon Footprint — 10 marks

5/5 answered

Q4 Multilevel Inheritance for Circular Economy — 10 marks

A circular-economy system classifies products through multiple levels such as ElectronicDevice → Smartphone → RecyclableSmartphone. Implement the hierarchy.

Your Code

```cpp
#include <iostream>
using namespace std;

class ElectronicDevice {
public:
    void device() {
        cout << "Product: Electronic Device" << endl;
    }
};

class Smartphone : public ElectronicDevice {
public:
    void phone() {
        cout << "Type: Smartphone" << endl;
    }
};
```

Input

stdin input

Output

Visible Test Case

Test Case 1

Snipping Tool

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5/5 answered

Q4 Multilevel Inheritance for Circular Economy — 10 marks

A circular-economy system classifies products through multiple levels such as ElectronicDevice → Smartphone → RecyclableSmartphone. Implement the hierarchy.

Your Code

```cpp
#include <iostream>
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class ElectronicDevice {
public:
    void device() {
        cout << "Product: Electronic Device" << endl;
    }
};

class Smartphone : public ElectronicDevice {
public:
    void phone() {
        cout << "Type: Smartphone" << endl;
    }
};
```

Input

stdin input

Output

Visible Test Case

Test Case 1

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- **Q4** Multilevel Inheritance for Circular Economy — 10 marks
- **Q5** Abstract Class for Carbon Footprint — 10 marks

5/5 answered

Q3 Function Overriding in Green Transport

10 marks

A green-transport platform calculates environmental impact differently for ElectricBus and DieselBus. Analyze how function overriding improves runtime behavior.

Your Code

```cpp
#include <iostream>
using namespace std;

class Bus {
public:
    virtual void environmentalImpact() {
        cout << "Bus environmental impact." << endl;
    }
};

class ElectricBus : public Bus {
public:
    void environmentalImpact() override {
        cout << "Electric Bus has low environmental impact." << endl;
    }
};
```

Input

stdin input

Output

Visible Test Cases

Test Case 1

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Q2 Hierarchical Inheritance for EVs

An EV company manages different vehicle types such as ElectricCar and HybridCar. Analyze how reuse common vehicle and battery properties.

Your Code

```cpp
#include <iostream>
using namespace std;

class Vehicle {
public:
    string type;
    int battery;

    void getData() {
        cin >> type >> battery;
    }

    void display() {
        cout << "Vehicle Type: " << type << endl;
        cout << "Battery Capacity: " << battery << " kWh" << endl;
    }
};
```